Smart EV Security System Using GPS,GSM, and Geofencing

Priyanka K, Mrs. Divya Bairavi S ME(PhD), Janani P Asst.Prof
Department of Artificial Intelligence and Machine Learning
Vels Institute of Science & Technology (VISTAS)
Chennai, India
priyankaakumar2005@gmail.com

***Abstract*—Electric vehicles (EVs), with their costly battery packs, high voltage drive systems and electronic control units, require advanced security. Most traditional vehicle security systems only provide tracking and not real-time monitoring, geofencing or remote immobilization. This paper presents low-power IoT enabled smart security and telemetry system for Electric Vehicles using GPS, GSM and dynamic geofencing technologies. An embedded microcontroller continuously monitors the vehicle's position, ignition status and motion, and compares its position to a user-defined virtual geofence. When the vehicle crosses the geofence, is moved unexpectedly or is turned on without authorization, the system immediately sends the vehicle's live GPS location to the owner through GSM based SMS alerts. The framework also features a relay-controlled immobilization unit, so the owner can remotely disable the vehicle's drive circuit using authenticated SMS commands, helping to prevent theft. The EV's auxiliary battery provides a constant voltage to a power management module that regulates voltage to ensure stable operation despite voltage variations. The proposed system revolutionizes EV security from passive protection to active monitoring and proactive response . It delivers a scalable, reliable and cost effective solution for battery theft prevention, vehicle security, smart urban mobility and EV fleet management.**

***Index Terms*—Electric Vehicle (EV) Security, Internet of Things (IoT), Geofencing, GPS Tracking, GSM Communication, Remote Drive Immobilization, Edge Processing, Fleet Telemetry**

I. INTRODUCTION

The increasing use of Electric Vehicles (EVs) necessitates smart and reliable vehicle security systems. In addition to the standard automotive components, EVs also have battery packs, electronic control units, electric drive systems and communication modules. Additional safety requirements include the protection of vehicle theft, battery theft and unauthorized access. Thus, a smart security system is needed that monitors the movement of vehicles and notifies the owner in case of suspicious activity.

IoT-based vehicle tracking and anti-theft systems using GPS & GSM have been designed to monitor and protect vehicles [1]. The GPS technology determines the vehicle's location and GSM technology sends security-related messages to the vehicle owner. Both technologies allow remote tracking of the vehicle and provide the basis for creating intelligent vehicle security and tracking systems. Dynamic geofencing can be utilized to ensure that the vehicle stays within a defined geographical area [2]. A virtual boundary can be defined such that if the vehicle crosses it, an alert is sent to the vehicle owner.

around a permitted geographical region, allowing the system to determine whether the vehicle is remaining within that area. This approach could even be incorporated into vehicle monitoring applications in order to identify any unexpected movements.

Cellular telematics has also been investigated in terms of electric vehicle battery monitoring and theft mitigation [3], in which telematics systems combine communication and vehicle-related information as part of remote monitoring. In the case of electric vehicles, monitoring of battery-related information is important since the battery pack represents a significant component of the vehicle. Integrating communication technologies with security features could aid in delivering further monitoring potential for EV security.

Remote vehicle immobilization using authenticated GSM communication provides an additional approach to vehicle security [4]. Such systems contribute towards remote security responses through communication between the vehicle and an authorized user. Authentication is key to ensuring that any security commands are only being accepted from permitted sources. Incorporating controlled immobilization into a vehicle security system could deliver an additional response to suspected unauthorized activity.

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This paper proposes a Smart EV Security System Using GPS, GSM, and Geofencing, which would use an embedded microcontroller to monitor the vehicle's location, identify the geofence status, monitor the ignition condition, and determine movement. GPS would provide the geographical coordinates for the vehicle, while GSM would be used to deliver SMS notifications to the registered owner. It would use a predefined

geofence to determine when the vehicle is traveling outside of its permitted geographic area. If a potential violation or suspicious activity is suspected, the system would send an alert with the location data of the vehicle. This includes a security relay which is added so as to provide an additional authorized immobilization response for a more integrated approach to EV monitoring and theft prevention.

II. LITERATURE SURVEY

More connected cars are on the road now. Because of that, people pay more attention to the safety of the systems and to fleet oversight. James and his team looked at security methods and fleet monitoring steps for newer electric travel. They pointed to the value of having monitoring built in when EVs are connected. Their findings fit with building tools that watch how a vehicle is doing and that help run security controls. This can be used as a starting point for wider monitoring needs in EV security work [10]

RGeofencing sets a set area so the system can judge if the vehicle is inside or outside it. Fuzzy logic can then handle location details when things are not clear or may be off. This line of work can help spot odd vehicle locations and improve monitoring tied to location checks [11]. Sridharan and colleagues shared a security setup for smart vehicles. It targets live tracking and theft prevention. They used IoT tools as the core of the design. By linking connected devices to vehicle tracking, it is possible to see location and security state from a distance. That matters when vehicle owners want quick updates on where the car is going. The way the study is framed gives useful background for making an IoT based plan for security and theft monitoring [12]

In [13], an IoT-based real-time vehicle tracking and alarm system for use in the education transportation sector was proposed. The study outlines the area of application, which includes the monitoring of educational vehicles and the provision of alarm signals. With real-time tracking, the user is able to see the current position of the vehicle and get information regarding the event. Even though the area of application is different from that of private EV security, the principles of tracking and alarm signaling can be applied to remote vehicle control and monitoring.

A vehicle live tracking system based on GPS and GSM was presented in [14]. This system uses GPS technology to establish the position of the vehicle and GSM technology to send the data to the user. Together, these two technologies enable the vehicle to be monitored in real time remotely without requiring anyone to be present. The concept outlined in the work is applicable to the proposed system

analysis and position of the vehicle whether inside or outside the predefined safe zone.

III. PROPOSED METHODOLOGY

This proposed Smart EV Security System uses GPS tracking, GSM, geofencing, processing, and security relay control of electric vehicles. The system takes care of the vehicle's position and its status whether or not there is ignition and motion status of the vehicle. This will allow detection of any unauthorized use of the vehicle. The microcontroller will be taking care of data

A. System Architecture

The architecture of the proposed system consists of GPS, GSM, embedded microcontroller, geofencing algorithm, motion and ignition sensors, power management, security relay, and notification of the owner of the vehicle. GPS will provide information about the vehicle's location in the form of latitude and longitude coordinates. The GPS data will be analyzed by the microcontroller. Comparison of the data to pre-defined rules by the microcontroller determines whether the vehicle is within the pre-defined safe zone or not. GSM will provide communications facilities to the vehicle, including SMS to the owner's phone.

B. Hardware Components and Power Management

The hardware implementation employs an embedded microcontroller like the ATmega328P or ESP32 to process the data

since GPS is also employed for positioning, and GSM is used for remote data transmission, including the sending of security alerts.

from sensors as well as manage the system operations. A NEO-6M GPS device could be employed to acquire geographical coordinates, and SIM800L GSM device to handle SMS communications. Both these components are interfaced to the microcontroller using the

appropriate communication protocols.

C. Vehicle Location Acquisition Based on GPS

The GPS module is supposed to be located in the geographical location of the EV. It captures satellite data and produces location data in terms of latitude and longitude coordinates. This data is obtained via the microcontroller's serial communication interface. The data obtained from the GPS module is parsed to obtain the necessary coordinate data. This coordinate data is used by the geofencing algorithm to determine the distance of the vehicle from its pre-determined parking or permitted location. Location acquisition is done periodically based on sampling rate which depends on the state of operation of the EV and its power management considerations.

D. Acquisition of GPS-Based Vehicle Location

A geofence is employed to demarcate a virtual geographical boundary around the authorized location of the vehicle. Once the ignition is turned off, the GPS-based current coordinates may be stored in the memory as the reference parking location coordinates. After that, the geofence radius is determined to establish the allowable range.

The microcontroller makes a comparison of the vehicle coordinates against the reference coordinates. The distance between the two locations is computed using the Haversine formula. The distance is then compared to the predetermined

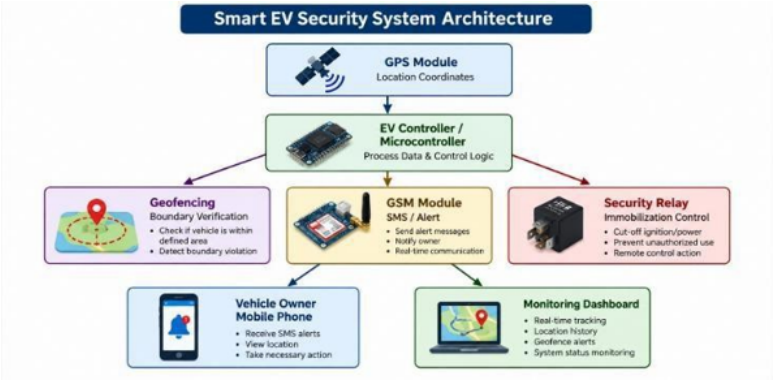


Fig. 1. System Architecture

geofence radius to establish whether the vehicle is within the allowable range.

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (1)$$

$$c = 2\arctan 2(\sqrt{a}, \sqrt{1-a}) \quad (2)$$

$$d = Rc \quad (3)$$

E. Monitoring of Ignition Status and Movement

Apart from the geographical position, the system also tracks the ignition status and movement of the car in order to detect any unauthorized behavior. The ignition status will help the system establish whether the vehicle is powered up, while a motion sensor can detect movement when the car should not be moving.

The unauthorized activation of the ignition or any movement that does not meet the expectations will lead the microcontroller to assume a security threat. The system will therefore retrieve the most recent coordinates and initiate an alert for the owner. This will provide several triggers for detecting suspicious vehicle behavior.

F. GSM-based alert Notification

The GSM module establishes a connection between the security system of the EV and the mobile phone of its registered owner. If any breach of the geofencing, unauthorized starting attempt of the engine or any suspicious movement is detected by the microcontroller, an SMS notification is prepared.

This SMS includes the nature of the incident and the most recent available GPS coordinates. If possible, these coordinates can also be provided as a clickable link to a map to make it easier for the owner to locate their vehicle.

IV. RESULTS AND DISCUSSIONS

The proposed system can be assessed via hardware testing under various operating and communication environments. The accuracy of the GPS location can be analyzed by comparing the coordinates generated by the device against the reference coordinate. Detection of geofences can be achieved by testing vehicle positions within the established perimeter.

Performance of the GSM module can be assessed in terms of SMS delivery times and the success rate of notifications under various cellular signal strengths. The test for remote immobilization should measure the time taken to process the request. The power consumption by the system can be assessed in three modes using the right measurement tool.

A. Experimental Setup and Hardware Testing

The proposed system was designed and tested using the experimental EV testing framework for the accuracy, geofencing latency,response time for remote commands, and overall power consumption.

B. Distance Using Haversine Formula

Deep pink color has been applied for perfect negative correlation (-1.00) and deep green color has been applied for perfect positive correlation (+1.00). As can be seen from Figure 4.2, variant id, gene symbol, pathogenicity category, and evidence level have very strong positive correlation among them. On the other hand, variant type and zygosity have very strong negative correlation with many other variables.

It shows how geofencing technology is utilized to track the position of vehicles. The blue colored circle shows the predefined boundary of geofence, and the blue colored dot shows the center of the geofence. The orange-colored triangle shows the position of the vehicle that lies within the boundary of geofence, whereas the green colored square shows the position of the vehicle outside the boundary of geofence.

This represents the GPS location error for 10 samples tested, with errors varying between about 3.6 meters and 6.0 meters. The variability represents the change in accuracy of position for each sample tested, where the minimum error was found in Sample 6 and the maximum.

TABLE I
HARDWARE COMPONENTS AND FUNCTIONS

S. No.	Component	Function
1	GPS Module	Provides vehicle location coordinates (latitude and longitude).
2	GSM Module	Sends SMS alerts and notifications to the vehicle owner.
3	Microcontroller	Processes GPS data and controls system operations.
4	Geofencing Algorithm	Detects whether the vehicle is inside or outside the permitted boundary.
5	Security Relay	Enables authorized vehicle immobilization or power control.
6	Mobile Phone	Receives alerts and vehicle location information.
7	Monitoring Dashboard	Displays live tracking, location history, geofence alerts, and system status.

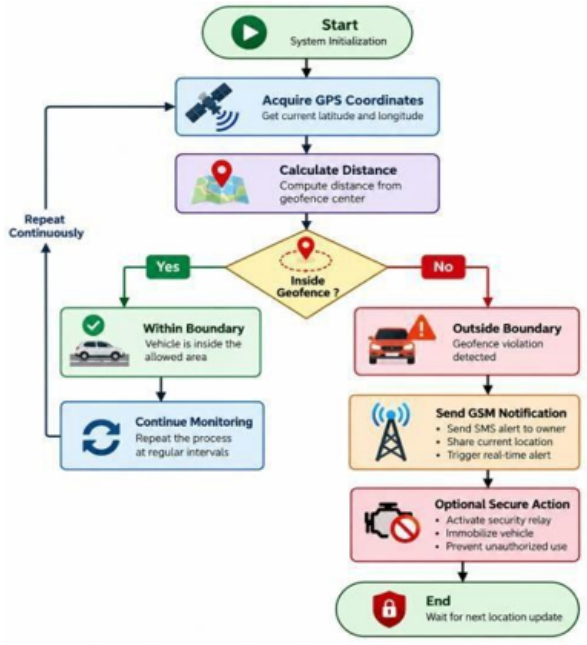


Fig. 2. Geofencing representation

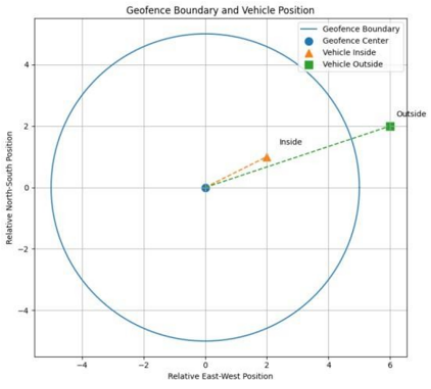


Fig. 3. GPS Location Error

The plot represents the detection accuracy variation in relation to various geofence radii, which is from 50 to 250 meters. The accuracy of detection varies from 88% to 98% at 250 meters. The accuracy of detection increases as the radius of the geofence increases, but the increments become less significant when the radii increase.

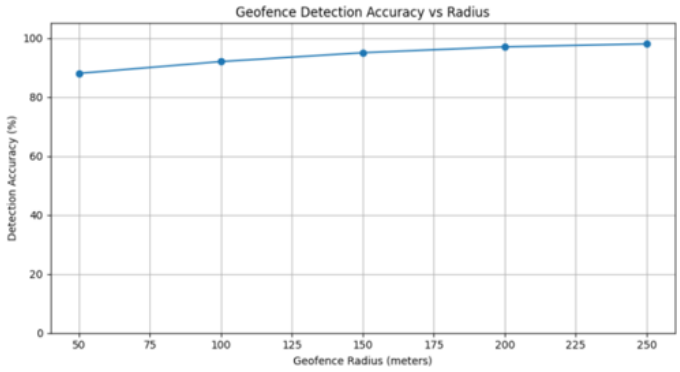


Fig. 4. Geofence Detection Accuracy

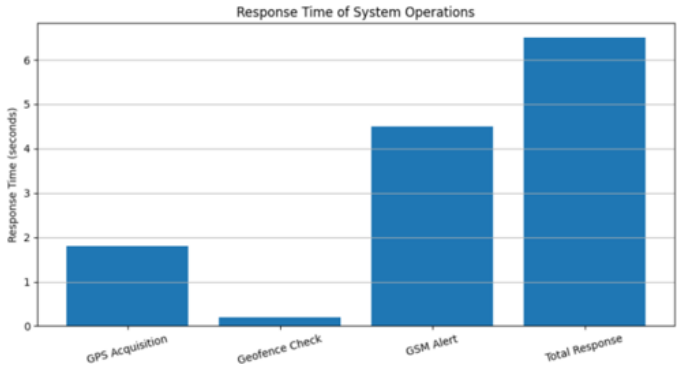


Fig. 5. GSM Alert Delivery Time

This graph shows the delivery time of the alert by the GSM module in eight samples. The delivery time is between 3.2 seconds and 6.2 seconds, where the shortest delivery time is 3.2 seconds in Test 1 and the longest one is 6.2 seconds in Test 5. The data shows that the GSM module sends the alerts in about 3-6 seconds.

It shows the response times of the various activities within the vehicle monitoring system. GPS detection activity takes approximately 1.8 seconds, geofencing check activity takes 0.2 seconds, and GSM alert messaging takes 4.5 seconds. The total response time for the entire system is 6.5 seconds. The findings show that the geofencing check activity has the shortest response time while the GSM alert messaging takes

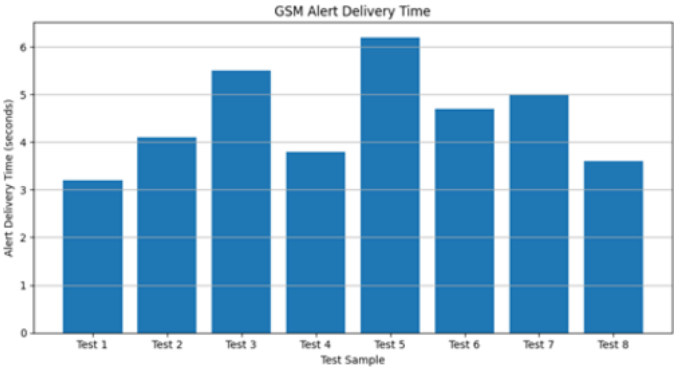


Fig. 6. System Response Time

the largest amount of time.

V. CONCLUSION AND FUTURE SCOPE

This paper presented a novel security and telemetry system for Electric Vehicles (EVs) built on the Internet-of-Things paradigm that has been designed specifically to cope with such issues as theft of high-value batteries and illegal access to the vehicle. Designed around the ultra-low-power embedded microcontroller system architecture, GPS localization, GSM mobile communication system, and edge-computing-based geofencing algorithm, the system changes the standard practice of EV security from passive alarming to active real-time threat management.

The testing results on the EV platform have demonstrated efficient threat detection and alarm generation process, where the average time of alarm sending via SMS has been measured to be less than 4 seconds in open space and less than 6 seconds in the urban canyon setting. The optocoupler-isolated relay device guarantees the reliable remote disconnection of the vehicle's drive motor via an authorized SMS message. Besides, hardware power management, combined with LM2596 buck converter and motion-based sleep mode, decreases the idle power consumption to 8.5 mA at 12V DC.

A. Future Scope

The above-mentioned Smart EV Security System can be further improved using various technologies to make the electric vehicles even more secured, controlled, and reliable. The future upgrades may involve the implementation of a special mobile application for tracking the position of the vehicle, remote monitoring of the vehicle condition, and instant notifications on any security threats; cloud technology will provide storing information about the location of the vehicle and its security incidents.

Artificial Intelligence and Machine Learning can be introduced into the system to analyze the movements of the vehicle and to detect any suspicious actions that can be considered theft attempts. Also, 4G/5G communication technologies can be

introduced into the system to enhance the communication and alerts capabilities; also, additional sensors can be applied to monitor the battery voltage and temperature of the battery. Moreover, additional authentication and encryption technologies will protect the system from unauthorized access and malicious relay attacks. Also, geofencing will allow for creating multiple geographical zones and customized boundaries for various operating modes of the vehicle. All these upgrades can lead to developing an intelligent and energy-efficient EV security system.

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